SAMPLE

Famous Astronomers: Edwin Hubble

The Expanding Universe

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Free sample: Introduction & Chapter 1

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Introduction

The man who told the twentieth century how big the universe is began his working life in a Kentucky law office he could not stand, and the distance between those two facts is one of the great comic openings in the history of science. This is the story of Edwin Hubble, the Missouri born astronomer who proved that the Milky Way is not the whole universe, that the spiral nebulae are galaxies of their own, and that those galaxies are running away from one another in a pattern the century read as creation's own expansion, the man who gave the expanding universe its observational footing and whose name now rides the space telescope, the law, and the constant that the modern cosmology still argues over.

The chapters tell the whole life: the athlete's youth and the Oxford polish, the telescopes and the plates, the night he resolved Andromeda, the decade he measured the galaxy retreat, the colleagues from the mule driver to the famous physicist, and the afterlife in which his name became the science's most famous brand.

The name deserves a line before the chapters start, for few scientists of the century became brands the way this one did: the space telescope the public knows by his name, the law and the constant the schoolbooks print, the crater and the asteroid the sky itself carries, and the standing irony that the man whose name is now the public's shorthand for the expanding universe distrusted the theory the expansion rode on, the fame the rest of the book inherits and the honest biography must keep in view, the book's own starting fact being that the name is larger than the life and the life is larger than the legend.

The reader should settle early what kind of chronicle this is, since the life divides into a public career and a private performance, and the two are not the same thing. The career is the observatory record: the papers, the counts, the law, the honors, the evidence the modern cosmology rests on. The performance is the persona the astronomer built and wore: the acquired accent that outlived the years that justified it, the tweeds and the pipe at the mountain observatory, the fencing of his own legend against all comers, the lawyer's habit of claiming priority that cost him good will and, once, a share of a discovery. The honest biography has to carry both, and the chapters do: the science chapters give the career its full due, and the human chapters give the performance its full reading, the catalog's standing habit of paying for the honesty up front.

The series' note closes the preview, the book's scope set out plainly: this volume holds the observer's whole life, the science and the persona, the

record and the performance, the era and the afterlife, and the reader who wants the theory's own account will find the Belgian's volume on the neighboring shelf, the catalog having given the prediction and the confirmation their own rooms, the preview's close carrying the series' own promise of the whole honest account.

The preview follows the life's own order: the origins and the education first, the Midwest youth and the Oxford years; the mountain second, the great reflectors and the observing craft; the galaxies third, the resolution of the island universe debate; the law fourth, the retreat of the galaxies and the expanding universe it announced; the peers fifth, the rivals and the collaborators the story is entangled with; the afterlife sixth, the corrected constant, the renamed law, and the space telescope that carries his name; and the pocket last, the athlete's records, the pipe's rituals, and the graves the story ends at. The chapters unpack each item in the catalog's standing sequence, the biography's honest whole.

A note on the account's two hazards, because this biography's subject carries two discoveries the textbooks print in bold, and a life told by its headlines can blur. The two headline words, galaxies and the law of redshifts, get their full stories in the science and discovery chapters, each with its own dates, instruments, and co-workers; the chapters between carry the rest of a crowded life, the athlete, the Rhodes scholar, the performer, the war officer, and the California institution builder, and the account keeps all of it in frame. The reader who wants the man as well as the measurements will find him here whole, and the reader who wants only the universe's enlargement can read the two bold chapters and leave satisfied, the book serving both appetites honestly.

One scope note on the volatile items, because a life this documented still has its soft places. The childhood anecdotes arrive through the family's own retellings and the memoirs of the widow, stories polished by decades of retelling; the persona's origins, the accent, the manner, are reconstructed from colleagues' letters and the era's gossip rather than the man's own account, he having left no self-portrait in words; and the record's famous silences, the unkept diary and the unwritten autobiography, are themselves part of the record, the facts the account flags rather than fills. The biography thus keeps its confidence where the paper is solid, the plates, the papers, the observatory's logbooks, and its humility where the sources are human.

Chapter 1: The Discovery of Edwin Hubble

The discovery story of a person is the story of how they came to be found by their life, and this astronomer's life found him slowly, through a youth of many

talents and a decade of deliberate detours.

The family's texture belongs in the origins' first room: the grandfather the boy was named for, the disciplined household the insurance man ran, the mother whose piety the son respected at a formal distance all his life, the brothers and sisters the career would later support and outgrow, and the standing family note the biographies print, the childhood illness the boy survived, the scar the youth carried and the headstrong constitution the man rode for six decades, the origins' honest room being the one the fact sheet dates and the pocket chapter's stories return to.

The high school's record belongs in the origins too, the pocket's earliest entry: the athlete's letters the local paper printed, the records the graduation class predicted, the prophecy the yearbook set down, the young man bound for greatness the joke the era's hindsight perfected, the entry being the biography's own opening exhibit, the pocket chapter returning to it as the first room of the legend the life became.

The origins come first. The future astronomer was born in a small Missouri river town in the autumn of eighteen eighty nine, the third of seven children of an insurance man whose work moved the family steadily westward, and the boy's schooling ran through the Midwest's public schools until the family settled in the Chicago suburbs in his early teens. The youth's record was the kind the local papers notice: the athlete who lettered in most of the sports a high school offered, the scholar who ran his own delivery route, the young man who tore through the county's records... the story the family told and the papers printed. The tall, strong teenager left for the university with a reputation already made, the athlete's body and the scholar's appetite arriving on the campus together.

The Oxford texture belongs in the second room's close: the rowing and the running the college's records kept, the boxing the gymnasium's visitors retold, the student theater the young American performed in, and the famous physical examination the Rhodes physicians gave him, the perfect specimen's score the family's letters retold for decades, the room's closing note being the England the young man built his adult self in, the accent and the tailoring and the unhurried manner all dated from these years, the persona's own birthplace the room records and the feature chapter tours.

The university years come second, and the two appetites ran in parallel. The Chicago undergraduate studied mathematics and astronomy under the university's pioneers, won his way into the physics laboratories of the famous experimentalist whose department he later claimed closer ties to than the record quite supports, and lettered in track, basketball, and boxing, the

heavyweight frame serving the science's later legend. The Rhodes scholarship took him to Oxford in the tradition of the era's brightest Midwesterners, and there the story takes its famous turn: the young American read law because his dying father had wished it, absorbed the English manner as thoroughly as the legal texts, rowed and boxed and perfected the accent, and came home in nineteen thirteen with a degree he would never use, a lawyer's training he would quietly turn into a scientist's rhetoric, and a persona he would wear for the rest of his life.

The detour years come third, and the biographies all give them the comic space they earn. The returned Oxonian opened a law office in the Kentucky city where his family had landed, drew up wills and contracts for the better part of a year, and hated every hour of it; the escape came through the classroom, a year of teaching Spanish and physics and coaching the high school's basketball team to the state's title game, and then the honest surrender, the return to the university for the doctorate the first career had interrupted. The doctoral work ran at the Wisconsin observatory of the university's satellite station, the thesis on the photographic study of the faint nebulae, the subject that would own the rest of his life, and the war took him from the defense straight into the uniform, the major's commission and the stateside service that his later telling would stretch toward the front lines the record never confirmed.

The doctoral trivia belongs in the room's close: the thesis's subject the faint nebulae were, the photographs the Wisconsin years took, the defense's timing the army's orders respected, the young doctor taking the examination in uniform and reporting for duty the same season, the close being the era's own compression, the scholar and the officer sharing one calendar year, the detour account's standing image of the life's discipline.

The war's honest record belongs in the detour's close: the commission the army gave the young doctor, the stateside instruction the service records show, the armistice's timing the legend stretched, and the major's rank the man kept as a title for life, the close's honest note being the biography's own correction, the combat the later telling implied and the records never confirmed, the room being the era's own measure of the man's need for the legend,

the detour chapter's standing honesty.

The mountain comes last in the discovery's order: the young doctor arrived at the southern California observatory in nineteen nineteen, within months of the great hundred inch reflector's own arrival, the two careers, the man and the machine, beginning together on the mountain that the decade's cosmology

would be written from, the discovery story's close being the standing irony that the century's greatest observational cosmologist spent his first career running from the science he was born for, and the science simply waited for him.

The Yerkes years earn a full room in the discovery chapter, because the detour's end came through a Wisconsin observatory and a telescope that was not large. The doctoral work, finished in the war's first years, photographed the faint nebulae, the smudges the era could not classify, and the thesis marked the young lawyer as a serious observer with a nose for the field's biggest open question; the director who hired him for it was the same astronomer whose observatory culture valued patient photography over grand theory, and the student's apprenticeship in long exposures began there, the craft the later discoveries would run on. The discovery chapter keeps the room honest: the man arrived at his life's work by way of a lake, a lawyer's detour, and a thesis on the very smudges he would later unmask.

The arrival's timing earns the second room, because the life's fortune was calendrical. The world's largest telescope, the hundred-inch mirror, saw first light in the year the United States entered the war; the young astronomer, finished at Yerkes and newly enlisted, crossed to France and came back when the demobilization released him; and the mountain hired him into the shadow of the instrument that had been waiting, hardly used, through the war years. The coincidence the chapter honors is the era's own: the biggest glass and the right observer came to the same mountain within two years, the war having delayed both, and the universe's map began to move the moment the pair met.

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